in the many-worlds interpretation can't work that way. With Schrödinger's cat, there are two possible outcomes—the cat is either alive or dead—so it's tempting to think the probability of either outcome is one in two: 50 percent. But say we had set up the experiment slightly differently—say we hadn't left the cat in there quite so long, out of some belated sense of mercy, and the probability of a radioactive decay (and thus a dead cat) was only 25 percent, rather than 50 percent. Now we have a problem: there are still two possible outcomes, but quantum physics dictates that they have unequal probabilities. There's a 75 percent chance that the cat is alive and a 25 percent chance that the cat is dead—but there's still just two branches, each inhabited by nearly identical versions of you. Is the copy of you in the dead-cat branch somehow "less real" than the copy of you in the living-cat branch? How do we make sense of this?

It gets worse. This is just one experiment, and it's a big universe out there—in fact, in any reasonable understanding of Everett's interpretation, there's an infinity of branches on the universal wave function. How can we make sense of probabilities when there's an infinite number of copies of ourselves? The only reason we can calculate probabilities for our dice throws is that we can count the number of possible outcomes. This approach doesn't work in an infinite multiverse, because the numbers involved are always infinite. If we want to know the number of branches in which a particular event occurs—say, the number of branches of the universal wave function where you're reading this book right now—the answer will always be infinite. And the number of branches in which you are *not* reading this book is also infinite. So what's the probability, in the multiverse, that a randomly chosen version of you would be reading some version of this book? What does the fraction infinity over infinity come out to? (Figure 11.1b). There are entire branches of mathematics devoted to dealing with infinity, and they tell us that such fractions can equal nearly anything at all—zero, some finite number, or even another infinity. So how do we handle this? How can we recover the fabulously accurate probabilistic predictions of quantum physics in the totally deterministic universe of the many-worlds interpretation? How do we measure the infinite fraction of infinite outcomes in which you're reading this book? And what can it even mean to talk about probability in a world where literally anything that is physically possible actually occurs somewhere?